

A new phase-field method for simulating γ' precipitation in multicomponent nickel-base superalloys

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Abstract

The phase-field method is used to develop a new technique to simulate γ' precipitation in multicomponent Ni-base superalloys. This technique generates the Gibbs free energy for both the γ and γ' phases from the single function formalism of both phases in the CALPHAD method without using conventional CALPHAD software. The Gibbs free energy surfaces for eight-element multicomponent systems are used to confirm that there is an accurate coupling with the CALPHAD method, because they compare favorably with calculated results from Thermo-Calc. In addition, γ' precipitation in a third-generation Ni-base superalloy containing eight components is simulated in one dimension with anti-phase effects of the γ' phase on the precipitate interactions. The simulation results reveal late-stage enrichments of Re and W in the γ phase during coalescence of the γ' precipitates.

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1. Introduction

Modeling microstructure evolution is important for efficient alloy design as well as for understanding underlying phenomena in phase transitions. The phase-field method has been used to predict γ' precipitation from a γ matrix in Ni-base superalloys [1–8]. These calculated results were in good agreement with experimental data. However, such previously reported calculations were focused on binary Ni–Al [1–6] or ternary systems, e.g. Ni–Al–Mo [7] and Ni–Al–Cr [8], even though commercial Ni-base superalloys often incorporate as many as 10 elements, such as Co, Cr, Mo, Re and Ru. Therefore, there have been few examples of the phase-field method being used to design commercial multicomponent Ni-base superalloys. In order to design such multicomponent superalloys efficiently, it is essential to understand the effects of the content of each element on microstructure evolution, such as the temporal change

of the morphology and size distribution of γ' precipitates, in addition to the phase instability attributed to the excessive alloying (which possibly degrades creep strength).

One reason for the difficulty with multicomponents associated with the phase-field method is the complexity of deriving a chemical free-energy function that depends on the element content in a multicomponent system. There have been some simulations coupling with the CALPHAD (calculation of phase diagrams) method to provide the chemical free energy for the phase-field method in the multicomponent order/disorder solid-state phase transitions [7–13]. In the CALPHAD method, the Gibbs energy function of an ordered phase (such as the γ' phase) is usually described using a sublattice model with element site fractions. This site fraction is not used as a variable in the phase-field method. Therefore, the degree of freedom of the element site fraction was reduced to pass thermodynamic information obtained from the CALPHAD method to the phase-field method. This was achieved as follows: (i) linking to a datafile that contains the Gibbs energy and equilibrium conditions calculated from the CALPHAD

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software [7,10]; (ii) approximating the Gibbs energy surface obtained from the CALPHAD software as a parabolic function [11,12]; or (iii) obtaining the driving force calculated using the Thermo-Calc software [8,13]. In these methods, the order parameter in the phase-field method was independently defined from the CALPHAD information. Although each of these three methods has advantages and disadvantages [9], the first two are difficult to extend to higher-component systems because of the difficulty of predicting diffusion paths to build a datafile for method (i) and the mathematical complexity associated with method (ii). For method (iii), it is necessary to develop a numerical code that interfaces the Thermo-Calc software via the TQ interface for γ' precipitation for higher-order systems. To the authors' knowledge, previously simulated γ' precipitation with precipitate interactions was limited to binary and ternary systems [1–8].

The authors have previously used a Gibbs energy single formalism for γ/γ' phases with a four-sublattice model for ternary Ni–Al–Cr [14] and quaternary Ni–Al–Re–Ta [15] systems. In these simulations, the order parameter of each element corresponded to long-range ordering in the γ' phase, because the order parameter was expressed using the element site fractions of a four-sublattice model in the CALPHAD method. This approach was based on a report describing a binary Ni–Al system [2]. Nevertheless, there was still a gap in the number of elements between these model alloys and the typical multicomponent superalloys that are of commercial interest. This paper develops a new technique for simulating γ' precipitation in eight-element Ni-base superalloy systems without using conventional CALPHAD software.

There have been only a few reports on the γ' precipitate interaction with anti-phase effects in commercial multicomponent systems. Even with careful experiments, it is difficult to observe such a time series of precipitate interactions by tracking one precipitate. There have been many atom probe analyses focused on the local equilibrium at γ and γ' interfaces, and the morphology and size of the γ' precipitates, e.g. Refs. [16–21]. However, the temporal change in concentration during precipitate interactions was not previously reported. In the current study, the temporal evolution of element concentration during coalescence of the γ' precipitates in a third-generation Ni-base superalloy system (i.e. TMS-75 [22]) is simulated and discussed.

2. Modeling

This paper focuses on an eight-element system, TMS-75 [22], which was developed at the National Institute for Materials Science (NIMS) in Japan; i.e. Ni–13.7Al–12.6Co–3.6Cr–1.3Mo–1.7Re–2.1Ta–2.0W (at.%). One characteristic of the present coupling method with the CALPHAD method for a multicomponent γ/γ' phase transition was obtaining the Gibbs free energy using a single function formalism of both phases in the CALPHAD

method without using the conventional CALPHAD software. The Gibbs energies of γ (face-centered cubic (fcc)) and γ' ($L1_2$) phases were described using a four-sublattice model [23,24]. Fig. 1 shows a unit cell of the $L1_2$ structure with four interpenetrating simple cubic sublattices numbered 1–4 in the figure. These sublattices can be an origin of the $L1_2$ structure. Therefore, one domain of the γ phase and four different domains of the γ' phase exist in the present γ/γ' microstructure.

The molar Gibbs energy for the γ phase with an fcc structure is written [23–25] as:

$$G^{\text{fcc}}(c_i) = \sum_i c_i G_i^{\text{fcc}} + RT \sum_i c_i \ln c_i + \sum_i \sum_{j>i} c_i c_j \times \sum_{n=0}^m {}^n L_{ij}^{\text{fcc}} (c_i - c_j)^n + \sum_i \sum_{j>i} \sum_{k>j} c_i c_j c_k L_{ijk}^{\text{fcc}} \quad (1)$$

where i, j and k can be Al, Co, Cr, Mo, Re, Ta, W and Ni. R is the gas constant and T is the temperature. c_i is a mole-fraction of element i . G_i^{fcc} is the molar Gibbs energy of pure element i with fcc structure. ${}^n L_{ij}^{\text{fcc}}$ and L_{ijk}^{fcc} are binary and ternary interaction parameters, respectively. For the γ' ($L1_2$) phase, the molar Gibbs energy can be described as the sum of an ordering contribution of the γ' phase and the Gibbs energy of the γ phase shown in Eq. (1) [23–25] as:

$$G(c_i, y_i^{(s)}) = G^{\text{fcc}}(c_i) + \Delta G^{\text{ord}}(y_i^{(s)}) \quad (2a)$$

$$= G^{\text{fcc}}(c_i) + \Delta G^{L1_2}(y_i^{(s)}) - \Delta G^{L1_2}(y_i^{(s)} = c_i) \quad (2b)$$

where $y_i^{(s)}$ is the site fraction of element i on the s th sublattice, i.e. $s = 1, 2, 3, 4$. The term $\Delta G^{\text{ord}}(y_i^{(s)})$ is the ordering energy of the γ' phase as a function of the site fraction. When all sublattices have the same site fractions, i.e. $y_i^{(s)} = c_i$ in Eq. (2a), the ordering energy is zero. $G(c_i, y_i^{(s)})$ is then equal to the Gibbs energy of the γ phase, $G^{\text{fcc}}(c_i)$. Thus, the Gibbs energies of the γ and γ' phases can be expressed in a single function of Eq. (2a). It should be noted that such a single continuous expression between the γ and γ' phases is useful for the description of the Gibbs energy of a two-phase system in the phase-field method [2,9]. The ordering energy in Eq. (2a) can be rewritten as Eq. (2b). The term $\Delta G^{L1_2}(y_i^{(s)})$ in Eq. (2b) is described [23] as:

$$\Delta G^{L1_2}(y_i^{(s)}) = \sum_i \sum_j \sum_k \sum_l y_i^{(1)} y_j^{(2)} y_k^{(3)} y_l^{(4)} \Delta G_{ijkl}^{L1_2} + \frac{RT}{4} \sum_s \sum_i y_i^{(s)} \ln(y_i^{(s)}) + \sum_s \sum_i \times \sum_{j>i} y_i^{(s)} y_j^{(s)} \sum_{n=0}^1 {}^n L_{ij}^{L1_2} (y_i^{(s)} - y_j^{(s)})^n \quad (3)$$

The term $\Delta G_{ijkl}^{L1_2}$ in Eq. (3) is expressed in terms of the interchange bond energies [23–25] as:

$$\Delta G_{ijkl}^{L1_2} = U_{ij} + U_{ik} + U_{il} + U_{jk} + U_{jl} + U_{kl} + a_{ijkl} \quad (4)$$

where a_{ijkl} is a correction variable for some compounds. In Eqs. (1) and (3), $(c_i - c_j)$ and $(y_i^{(s)} - y_j^{(s)})$ take the elements in alphabetical order of their chemical symbol, i.e. Al, Co,

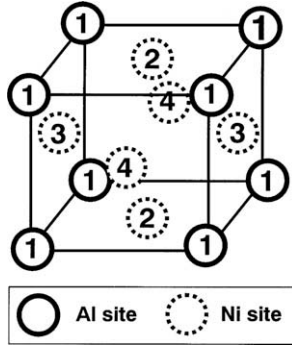


Fig. 1. A unit cell of L1₂ structure. Four interpenetrating simple cubic sublattices are numbered.

Cr, Mo, Ni, Re, Ta and W. All thermodynamic parameters (i.e. G_i^{fcc} , ${}^nL_{i,j}^{\text{fcc}}$, $L_{i,j,k}^{\text{fcc}}$, ${}^nL_{i,j}^{\text{L1}_2}$, U_{ij} and a_{ijkl}) were obtained from the thermodynamic NIST-Ni database [26,27].

Three order parameters were described using the element site fractions at four sublattices for a binary system [2,28,29]. In this study, the order parameters for each constituent element in the eight-element system were introduced because it was demonstrated that ordering behavior in a γ' phase depends on the constituent elements [30–33]. The order parameters of each element were defined using the element site fraction in the four-sublattice model:

$$\phi_i^1 = (-y_i^{(1)} - y_i^{(2)} + y_i^{(3)} + y_i^{(4)})/4c_i \quad (5a)$$

$$\phi_i^2 = (-y_i^{(1)} + y_i^{(2)} - y_i^{(3)} + y_i^{(4)})/4c_i \quad (5b)$$

$$\phi_i^3 = (y_i^{(1)} - y_i^{(2)} - y_i^{(3)} + y_i^{(4)})/4c_i \quad (5c)$$

$$c_i = (y_i^{(1)} + y_i^{(2)} + y_i^{(3)} + y_i^{(4)})/4 \quad (5d)$$

where $i = \text{Al, Co, Cr, Mo, Re, Ta, W}$. Eq. (5) can be rewritten as:

$$y_i^{(1)} = c_i(1 - \phi_i^1 - \phi_i^2 + \phi_i^3) \quad (6a)$$

$$y_i^{(2)} = c_i(1 - \phi_i^1 + \phi_i^2 - \phi_i^3) \quad (6b)$$

$$y_i^{(3)} = c_i(1 + \phi_i^1 - \phi_i^2 - \phi_i^3) \quad (6c)$$

$$y_i^{(4)} = c_i(1 + \phi_i^1 + \phi_i^2 + \phi_i^3) \quad (6d)$$

As for Ni:

$$c_{\text{Ni}} = 1 - \sum_i c_i \quad (7a)$$

$$y_{\text{Ni}}^{(s)} = 1 - \sum_i y_i^{(s)} \quad (7b)$$

where $i = \text{Al, Co, Cr, Mo, Re, Ta, W}$; $s = 1, 2, 3, 4$. Substituting Eqs. (1), (3), (4), (6) and (7) into (2b), the Gibbs energy of the γ and γ' phases was described as a single function of element mole fractions c_i and order parameters ϕ_i^j for Al, Co, Cr, Mo, Re, Ta and W.

Substitution of Eqs. (6) and (7b) into the first term of Eq. (3) is especially complicated for higher-component systems. The parameter $\Delta G_{ijkl}^{\text{L1}_2}$ in the first term of Eq. (3) includes six terms on U and one term on a_{ijkl} , as shown

in Eq. (4). The description of the terms on U can be rewritten as:

$$\begin{aligned} & \sum_i \sum_j \sum_k \sum_l y_i^{(1)} y_j^{(2)} y_k^{(3)} y_l^{(4)} \Delta G_{ijkl}^{\text{L1}_2}(U) \\ &= 4 \left\{ \sum_i U_{i,\text{Ni}} c_i^2 \sum_j \phi_i^j - \sum_i \sum_j (U_{ij} - U_{i,\text{Ni}} - U_{j,\text{Ni}}) \sum_k \phi_i^k \phi_j^k \right\} \end{aligned} \quad (8)$$

The total chemical free energy was defined as the sum of the local chemical free energy density and the interface energy [34] as:

$$F \equiv \frac{1}{V_m} \int_V \left(G + \frac{\alpha}{2} \sum_j (\nabla c_j)^2 + \frac{\beta}{2} \sum_j \sum_{i=1}^3 (\nabla \phi_j^i)^2 \right) dV \quad (9)$$

where G is the molar Gibbs energy of Eq. (2) as a single function of element mole fractions c_i and order parameters ϕ_i^j . The term V_m is the molar volume, which was assumed to be constant, $V_m = 7.0 \times 10^{-6} \text{ m}^3/\text{mol}$. The terms α and β are the gradient energy coefficients for the composition and order parameters, respectively. The values: $\alpha = 1.56 \times 10^{-14} \text{ J m}^2/\text{mol}$ and $\beta = 2.50 \times 10^{-16} \text{ J m}^2/\text{mol}$ were chosen so that the inter-phase interfacial diffuse chemical profile is $\sim 3 \text{ nm}$. This value was observed using both atom probe analyses [35,36] and the cluster variation method [33] for multicomponent systems. In this way, the interfacial energy is in the reported range of coherent solid–solid inter-phase boundary energies (i.e. 5–200 mJ m^{-2} [37]) because very few interfacial energy data for multicomponent systems have been previously reported. The interfacial energy of 65 mJ m^{-2} was obtained using the above gradient energy coefficients. For simplicity in this study, the composition-dependent elastic energy stemming from elastic inhomogeneity was not considered.

The temporal evolutions of element concentrations and order parameters were obtained by solving the following Cahn–Hilliard and Ginzburg–Landau equations:

$$\frac{\partial c_i}{\partial t} = \sum_j \nabla \cdot \left(\tilde{M}_{ij} \nabla \frac{\delta F}{\delta c_j} \right), \quad i, j = \text{Al, Co, Cr, Mo, Re, Ta, W} \quad (10)$$

$$\frac{\partial \phi_i^j}{\partial t} = -L \frac{\delta F}{\delta \phi_i^j}, \quad i = \text{Al, Co, Cr, Mo, Re, Ta, W}, j = 1, 2, 3 \quad (11)$$

where $\tilde{M}_{ij}$ is the so-called chemical mobility, with Ni as the dependent specie, and L is the structural relaxation coefficient. The chemical mobility $\tilde{M}_{ij}$ for the γ phase is related to the atomic mobilities of the constituent elements by:

$$\tilde{M}_{ij} = \sum_l (\delta_{il} - c_i)(\delta_{jl} - c_j) c_l M_l^{\text{Ni}} \quad (12)$$

where δ_{il} and δ_{jl} are Kronecker deltas, and the NIST mobility database [38] was used for M_l^{Ni} . Parameter L was determined from $\tilde{M}_{\text{AlAl}} = La_0/16$ [39] with the lattice parameter $a_0 = 3.35 \times 10^{-10} \text{ m}$.

Based on this definition, the interface migration velocity was controlled mainly by diffusion mobility. A semi-implicit Fourier spectral method [40] was employed for numerical analysis of the present one-dimensional system with a periodic boundary condition. The grid number and grid size were set to 256 and 0.5 nm, respectively. Nuclei were initially set randomly according to a Gaussian size distribution with an average of 3 nm diameter in the field. The initial concentration of each element was set to the nominal concentration of TMS-75 for all grid points, i.e. mole fractions of Al = 0.137, Co = 0.126, Cr = 0.036, Mo = 0.013, Re = 0.017, Ta = 0.021 and W = 0.02. In the nucleus regions, the values of equilibrium order parameters for γ' phases were set, i.e. Al = 0.953, Co = -0.305, Cr = 0.738, Mo = 0.930, Re = 0.082, Ta = 0.997, W = 0.561 and assumed to be zero in the other regions. The initial abundance ratio of four domains in the γ' phase was provided equivalently according to a random number.

3. Results and discussion

3.1. Validation of the Gibbs energy function

The Gibbs energy function as a function of order parameters and element content, i.e. Eqs. (1)–(4), (6) and (7), is temperature dependent, as shown in Eqs. (1) and (3). To validate the accuracy of the present coupling method with the CALPHAD method at arbitrary temperatures, the Gibbs free energy surfaces for γ and γ' phases in eight-element systems were compared with the calculated results from Thermo-Calc software [41] at 1100 °C in

Fig. 2a and 900 °C in Fig. 2b. The NIST thermodynamic database [26,27] was used for both calculations. Fig. 2a shows the Gibbs energy near the nominal composition of TMS-75 at 1100 °C. The mole fraction of Al was increased from 0.1 to 0.17 with constant mole fractions of Co = 0.126, Cr = 0.036, Mo = 0.013, Re = 0.017, Ta = 0.021 and W = 0.020. Fig. 2b shows the Gibbs energy as a function of the γ' volume fraction in TMS-75 at 900 °C. The present results were obtained under the condition of $\partial G / \partial \phi_i^j = 0$ ($i = \text{Al, Co, Cr, Mo, Re, Ta, W; } j = 1, 2, 3$) using the Newton–Raphson method. As defined in Eq. (2), the Gibbs energy of the γ' phase is described as the sum of the ordering energy of the γ' phase and the Gibbs energy of the γ phase. In these figures, the Gibbs energy of the γ' phase is shown only when the ordering energy is lower than or equal to zero. It is clear from Fig. 2 that both computed results show good agreement in Gibbs energy surfaces of γ and γ' phases.

It was demonstrated in this study that the Gibbs free energy surface described by the CALPHAD method can be obtained at arbitrary temperatures and compositions in the γ/γ' two-phase region for the phase-field method. Therefore, it was not necessary to construct a free energy function of a pseudo-binary system, to solve for chemical equilibrium conditions as was done in the KKS model [9,42,43] or to build a datafile of thermodynamic data for such a high-component system. However, in the present method, the formulation of the ordering energy term $\Delta G^{\text{L}_{12}}(y_i^{(s)})$ shown in Eq. (3) is complicated. In addition, for simplicity, the contribution of magnetic effects to the Gibbs energy was not considered here, although this contribution is small, as shown in Fig. 2, where there is good agreement with the calculated results from Thermo-Calc which included the magnetic contribution. Nevertheless, it was shown for the phase-field method that the Gibbs energy surface in eight-element, Ni-base superalloy systems was obtained by coupling with the CALPHAD method.

3.2. Order parameters linked to the element-site fraction in the CALPHAD method

As defined in Eq. (5), the order parameters were expressed using the element site fraction of the four-sublattice model in the CALPHAD method. Therefore, the order parameter has the meaning of long-range ordering of each element in the γ' phase. Fig. 3 shows the equilibrium order parameters ϕ_i^{eq} at 1100 °C, where $\phi_i^{\text{eq}} = \phi_i^j$. This result was obtained as a byproduct of the calculation in Fig. 2a under $\partial G / \partial \phi_i^j = 0$. The closer the value is to 1 or -1, the more the element substitutes for an Al site or a Ni site, respectively. When the value is zero, the element equivalently occupies both Al and Ni sites. These are understandable from the definition of Eq. (5), because values of three Ni site fractions out of four site fractions are identical in the L_{12} structure. Fig. 3 shows that the equilibrium order parameters of Ta, Al and Mo increased slightly with increasing Al concentra-

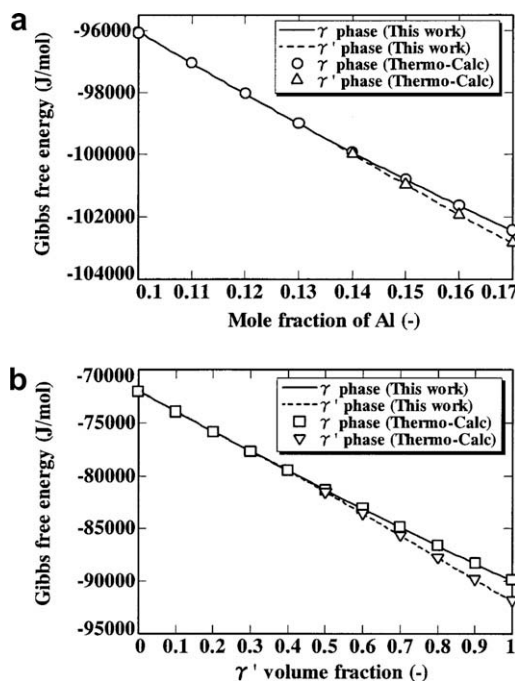


Fig. 2. Comparisons of the Gibbs free energy obtained in the present work with calculated results from Thermo-Calc in eight-element, Ni-base superalloy systems at (a) 1100 °C and (b) at 900 °C.

tion, while those of Re and Co decreased in this system. These results indicate that site fractions of Ta, Al and Mo at an Al site increased with increasing Al concentration, while Re and Co fractions at a Ni site increased. Such partitioning behavior differences between Al and Ni sites are characterized by a thermodynamic assessment in the CALPHAD method. In the present phase-field method, the temporal evolution of the order parameters linked to the element site fractions proceeds according to the Gibbs free energy surface obtained from the CALPHAD method.

3.3. Equilibrium interfacial profiles

Fig. 4 shows profiles of equilibrium element concentrations and equilibrium order parameters at the γ/γ' interface of TMS-75 at 1100 °C. These results were obtained from a single γ' precipitation calculation using the present phase-field method described above. The equilibrium order parameters of the inside γ' phase in Fig. 4c were Al = 0.953, Co = -0.305, Cr = 0.738, Mo = 0.930, Re = 0.082, Ta = 0.997 and W = 0.561. Among the seven elements, only Co showed a negative value of the order parameter, which means that Co preferentially substitutes for a Ni site in the γ' phase, as mentioned in Section 3.2. The interface width and the interface position between γ/γ' phases for concentration and order parameters depended on constituent elements. The order parameter of Ta inside the γ' phase was very close to 1, which indicates the strong tendency of Ta to occupy an Al site. Indeed, Ta showed a significant occupation of Al sites even at, and adjacent to, the inter-phase interface in the γ phase, as shown by the curve labeled A in Fig. 4. While the order parameter of Ta resulted in the preferential occupation of an Al site near the interface even in the γ phase, the average concentration of the lattice sites was the same as that of the bulk of the γ phase at these sites. This observation indicates the difference in the interface width and position between the chemical and structural profiles. In contrast, Mo, Cr and W showed a wide inter-phase interface in the γ' phase. For

example, this result is shown for W by the curve labeled B in Fig. 4. These observations suggest that the ordering degrees for Mo, Cr and W in the γ' phase adjacent to the interface are not as large as in the bulk of the γ' phase, although the average concentration of lattice sites is the same as the bulk. These profiles are characterized by the Gibbs free energy surface obtained by the CALPHAD method.

Table 1 shows a comparison of the equilibrium concentrations of the γ/γ' phases and the volume fraction of the γ' phase between those obtained in the present study, calculated results from Thermo-Calc, and experimental results [22,44]. The NIST thermodynamic database [26,27] was used for both calculations. In the microstructure of TMS-75, Al and Ta are rich in the γ' phase, whereas Co, Cr, Mo, Re and W are rich in the γ phase. Calculated phase compositions obtained from both the phase-field method and Thermo-Calc were almost identical. In the phase-field method, the volume fraction can be calculated according to

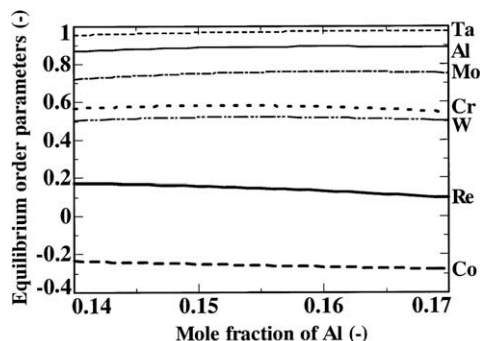


Fig. 3. Equilibrium order parameters of constituent elements at 1100 °C. The mole fraction of Al was increased from 0.14 to 0.17 with constant mole fractions Co = 0.126, Cr = 0.036, Mo = 0.013, Re = 0.017, Ta = 0.021, W = 0.020.

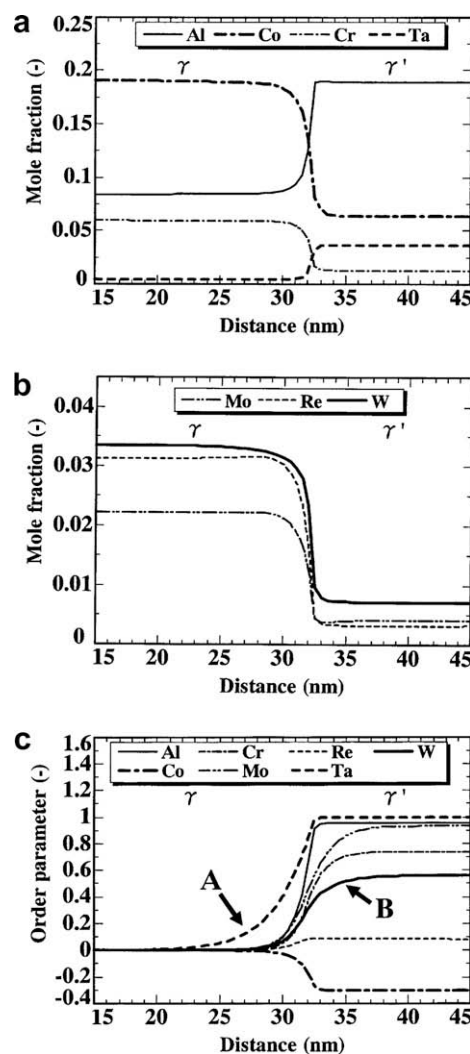


Fig. 4. Equilibrium concentration profiles of (a) Al, Co, Cr, Ta, (b) Mo, Re, W and (c) equilibrium order parameter profiles of constituent elements at and adjacent to the γ/γ' interface.

Table 1

A comparison of the γ and γ' phase compositions and the γ' volume fraction of TMS-75 at 1100 °C.

	Co	Cr	Mo	W	Al	Ta	Re	V_f
Phase-field calculation (this work)								
γ'	6.3	1.2	0.4	0.7	18.9	3.6	0.3	0.51
γ	19.1	5.9	2.2	3.3	8.4	0.4	3.1	
Thermo-Calc calculation								
γ'	6.3	1.3	0.4	0.7	18.9	3.5	0.3	0.51
γ	19.1	6.0	2.2	3.4	8.4	0.4	3.1	
Observation by Yokokawa et al. [44]								
γ'	10.0	1.8	0.7	1.6	16.9	2.7	0.4	0.55 [22]
γ	18.2	6.4	1.9	2.2	8.9	1.0	3.1	

a lever rule using the average concentration of all grid points. The total concentration of each element in the system was kept constant during the phase-field calculation, because the difference between the initial nominal concentration and the average concentration at the last stage was less than a mole fraction of 0.0003 for all elements, i.e. negligibly small. Therefore, the volume fraction obtained from the phase-field method was the same as the results obtained using Thermo-Calc. In regard to the comparison between the phase-field method and experimental results, the calculated Co concentration of the γ' phase was lower than the experimental result. In addition, W slightly more partitioned to the γ phase in the calculation. According to Murakumo et al. [22], the observed γ' volume fraction was ~ 0.55 at 1100 °C. However, the phase-field method predicted 0.51, which was reasonably close to the observation. These comparisons suggest that the predictions using the phase-field method are in good agreement with results of both Thermo-Calc calculations and experimental observations regarding the phase compositions and the volume fraction of the γ' phase.

3.4. γ' Precipitation in TMS-75

Fig. 5 shows snapshots of the computed temporal evolution of element concentration profiles during γ' precipitation of TMS-75 at 1100 °C. At the initial stage after nucleation, Al and Ta were depleted in narrow γ channels, shown by arrows C and D in Fig. 5a. γ formers, such as Co, Cr, Mo, Re and W, were saturated in the narrow γ channels. Saturation of Co and Cr was obvious in Fig. 5a, whereas Re and W were late in Fig. 5b. This delay in the saturation of Re and W was a result of the small diffusivity of Re and W. Local equilibrium was observed at the γ/γ' interfaces during the growth of precipitates. In this simulation, the development of “bow waves” of enriched solute ahead of the growing γ' phase during the diffusion of γ formers into the γ phase was confirmed. This “bow wave” was observed using an atom-probe analysis and was reported by Warren et al. [18]. The diffusion of Re and W into the γ' phase was significantly slower than for other elements, as indicated by arrows E and F in Fig. 5b. Subsequently, the element concentrations of the γ and γ' phases

almost reached equilibrium with uniformity, as shown in Fig. 5c.

At the late stage shown in Fig. 5d, γ' precipitates of I and J were merging across the γ phase at G. Such coalescence reduces the total energy of the system owing to the disappearance of interfaces [45]. It should be noted here that the precipitates H and I were also close to each other. However, they did not merge because the domains of these two precipitates were different, i.e. anti-phase domains. Fig. 6 shows profiles of the order parameter of Al in Fig. 5c. Four different domains in $L1_2$ structure can be distinguished by three-order parameters [28]. The four domains are $\phi_i^{\text{eq}}(-1, -1, 1)$, $\phi_i^{\text{eq}}(-1, 1, -1)$, $\phi_i^{\text{eq}}(1, -1, -1)$ and $\phi_i^{\text{eq}}(1, 1, 1)$. Order parameters for both I and J showed the domain of $\phi_i^{\text{eq}}(1, -1, -1)$. However, the domains of H and I were different, as H was $\phi_i^{\text{eq}}(1, 1, 1)$. The anti-phase boundary energy is generally high, which is revealed in this study by gradient energies of the order parameters and chemical concentrations [45,46]. It is therefore concluded that the anti-phase boundary of H and I was not formed in this system.

During the coalescence of the precipitates of I and J, Re concentration in the γ phase showed a pileup at position G, as shown in Fig. 5d. The concentration of Re was higher than the equilibrium concentration of the γ phase. After coalescence, a hump of concentration was confirmed for each γ former at position G in Fig. 5e. The humps ultimately disappeared in Fig. 5f. Fig. 7 shows the temporal concentration change of each element at G during coalescence of the precipitates of I and J. Elements Al, Co, Cr, Mo and Ta reached γ' composition early. However, concentrations of Re and W (with small diffusivity) gradually increased and were kept almost constant, respectively, until 1503 s. They then decreased to the γ' concentration. At 1503 s, Re and W were rich in the γ phase because other element concentrations had already transited from γ to γ' . The element concentration at 1503 s was mole fractions of Al = 0.098, Co = 0.16, Cr = 0.048, Mo = 0.016, Re = 0.036, Ta = 0.005 and W = 0.032. This result indicates that the coalescence of the precipitates at a late stage possibly produces a γ channel with high Re and W concentrations. Unfortunately, experimental detection of the temporal evolution of element concentrations during the coalescence of precipitates is generally difficult.

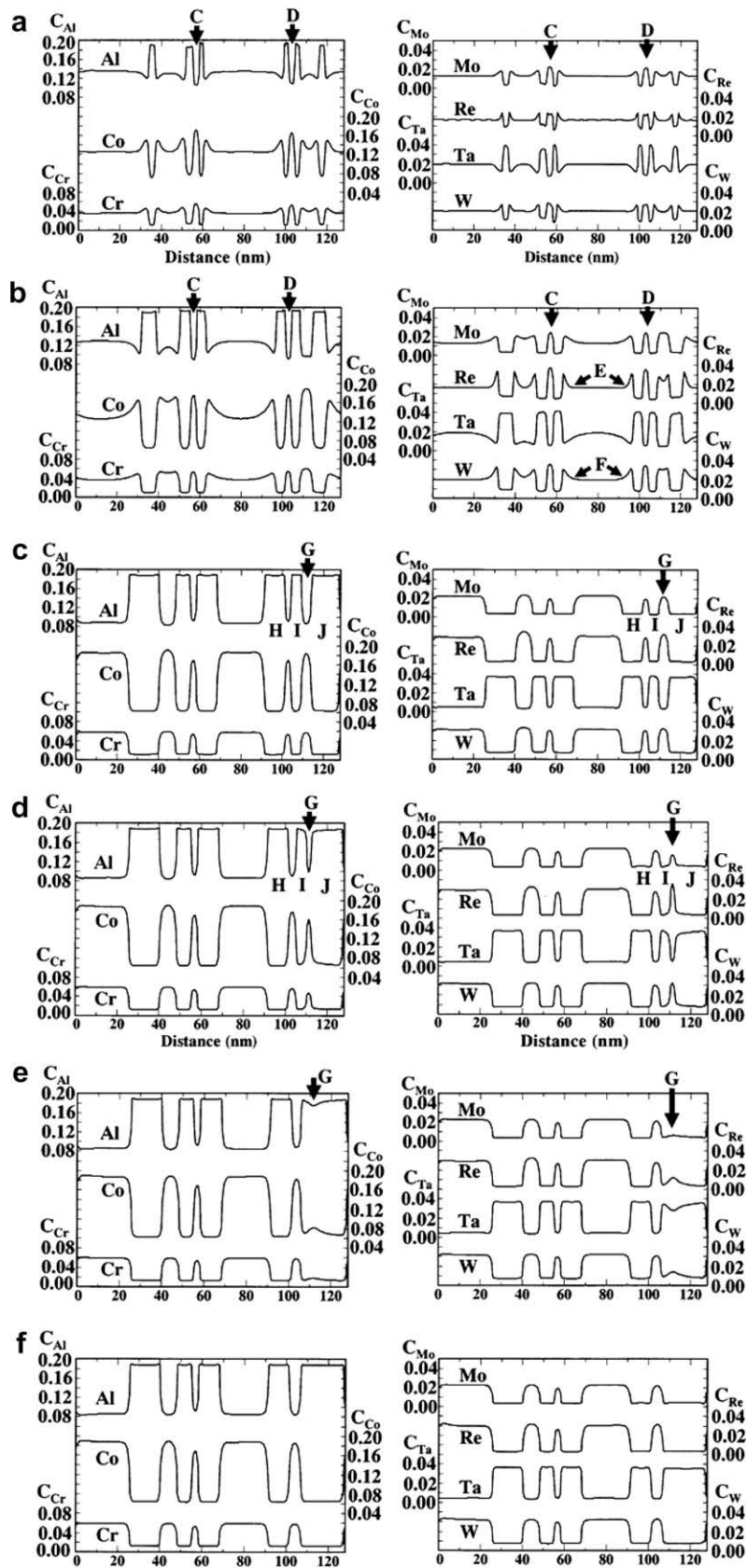


Fig. 5. Calculated temporal evolution of constituent element composition profiles in TMS-75 at 1100 °C: (a) $t = 11$ s; (b) $t = 28$ s; (c) $t = 1190$ s; (d) $t = 1503$ s; (e) $t = 1531$ s; (f) $t = 1701$ s.

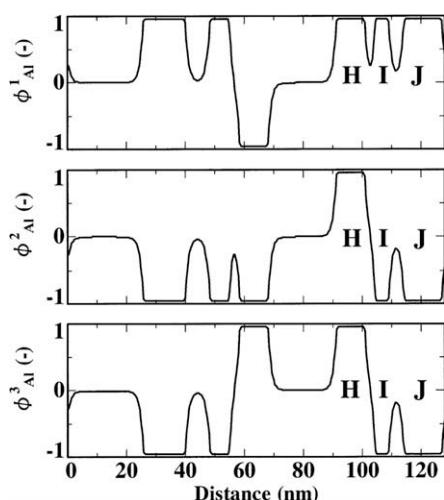


Fig. 6. Profiles of order parameters of Al in Fig. 5c.

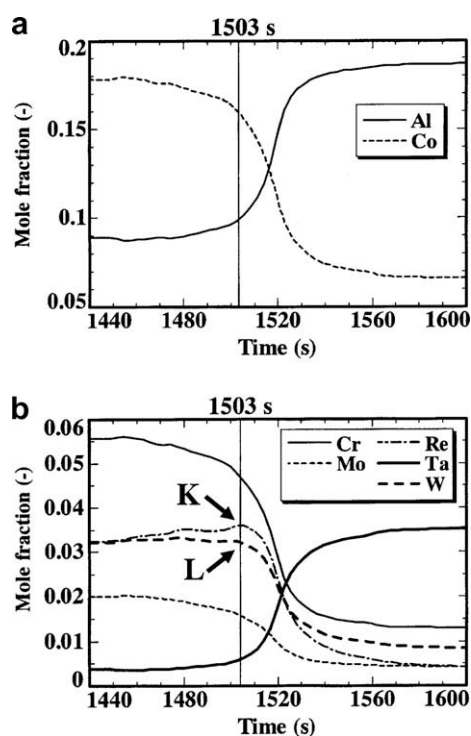


Fig. 7. Temporal change of element concentrations at position G in Fig. 5c during coalescence of γ' precipitates: (a) Al, Co; (b) Cr, Mo, Re, Ta and W.

4. Summary

A new multicomponent phase-field method was developed which is useful for simulating γ' precipitation from a γ matrix in multicomponent Ni-base superalloys. The Gibbs energy of the γ and γ' phases in eight-element systems was successfully obtained by employing the CALPHAD method without using conventional CALPHAD software. The order parameter of each element in the alloy was expressed using its element site fraction in a four-sub-

lattice model. It was therefore possible to distinguish four different domains of the γ' phase in a multicomponent microstructure. This capability also allowed anti-phase effects to be accounted for. The multicomponent phase-field method was further used to simulate γ' precipitation in TMS-75, a third-generation Ni-base superalloy. This method demonstrated both diffusion-controlled γ' growth under local equilibrium at the γ/γ' interfaces and precipitation interactions (such as coalescence). The simulated temporal evolution of element concentrations during γ' coalescence suggested that Re- and W-enriched γ channels could be formed during late-stage microstructure evolution.

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